

Instruction Manual

for J200-640 Continuous Zoom MW Infrared

Lens

1. Features

- a. Continuous zoom (including focus follow), manual focus, lens temperature feedback, zoom speed and focus step adjustable.
- b. Data calibration: the motor control board should be able to complete the calibration data and storage.
- c. The zooming process is smooth, and there is no obvious inflection point at the cam curve.
- d. No obvious cold reflection, even surface view of the black body.

2. Technical Specifications

- a. Focal length: 20mm ($\pm 5\%$) ~ 200mm ($\pm 5\%$) continuous zoom.
- b. FOV: $26.99^\circ \times 21.73^\circ$ ($\pm 5\%$) ~ $2.75^\circ \times 2.20^\circ$ ($\pm 5\%$) continuous zoom.
- c. F #: 4 (guaranteed by design).
- d. Transmission rate: $\geq 83\%$ (guaranteed by design).
- e. Waveband: $3.7\mu\text{m} \sim 4.8\mu\text{m}$.
- f. Adaptable detector: $640 \times 512 @ 15\mu\text{m}$, distance from the aperture to plane $19.1 \pm 0.1\text{mm}$.
- g. Optical axis consistency: deviation ≤ 1 pixel (Aligned with NFOV, then switch to WFOV), with whole process of zoom deviation ≤ 3 pixels.
- h. Repeatability error of optical axis in NFOV: ≤ 1 pixel.
- i. FOV switching time: $\leq 5\text{s}$ (under high and low temperatures);
- j. Protection: the first lens is coated with AR film.

3. Environmental Adaptability

- a. Low-temperature work: working temperature -40°C , maintain the temperature for 1 hour, and then work for 20min.
- b. High-temperature work: working temperature $+65^{\circ}\text{C}$, maintain the temperature for 1 hour, and then work for 20min.
- c. Low-temperature storage (testing along with the system): storage temperature -55°C , insulation 24 hours, and then test after returns to room temperature.
- d. High-temperature storage (testing along with the system): storage temperature $+70^{\circ}\text{C}$, insulation 48 hours, and then test after returns to room temperature.
- e. Vibration: Frequency range 5Hz~5.5Hz, amplitude 25.4mm; 5.5Hz~30Hz acceleration 1.5g; 30Hz~50Hz, amplitude 0.84mm; Acceleration from 50Hz to 500Hz is 4.2g. Three axis scanning, with two cycles per axis, each cycle lasting 15 minutes.

4. Interfaces

4.1 Optical interface

- a. Dimensions: The lens size is $(85 \pm 1) \text{ mm} \times (87 \pm 1) \text{ mm} \times (72 \pm 1) \text{ mm}$ (L $\times$ W $\times$ H), excluding cables, as shown in Figure 1.

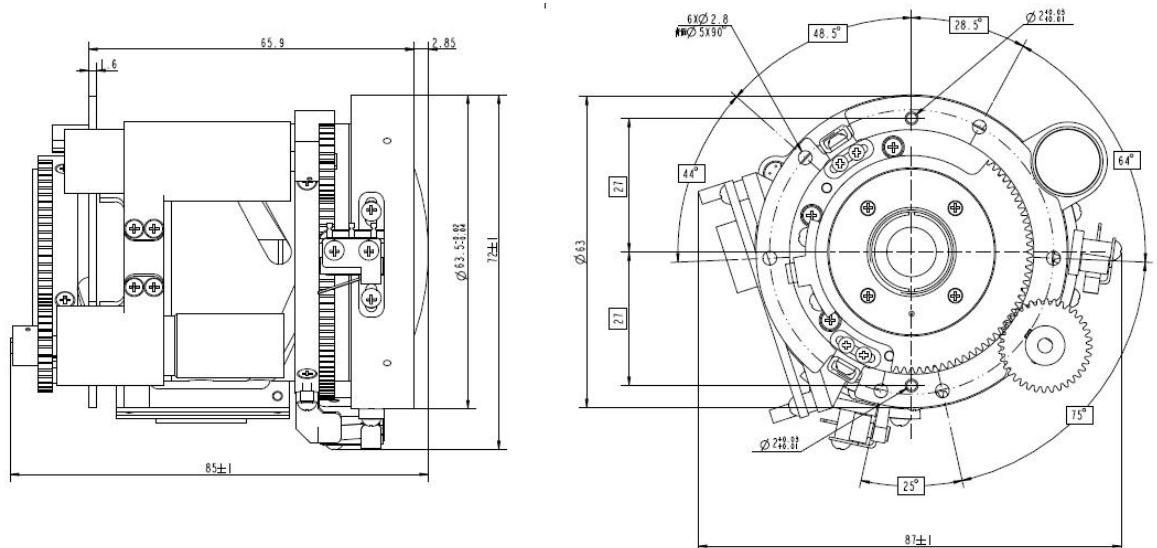


Figure 1 Dimensions of the lens

- b. Installation interface: The lens is fixedly installed with the system through the six $\phi 2.5$ countersunk through holes and two $\phi 2$ pin holes on the flange shown in

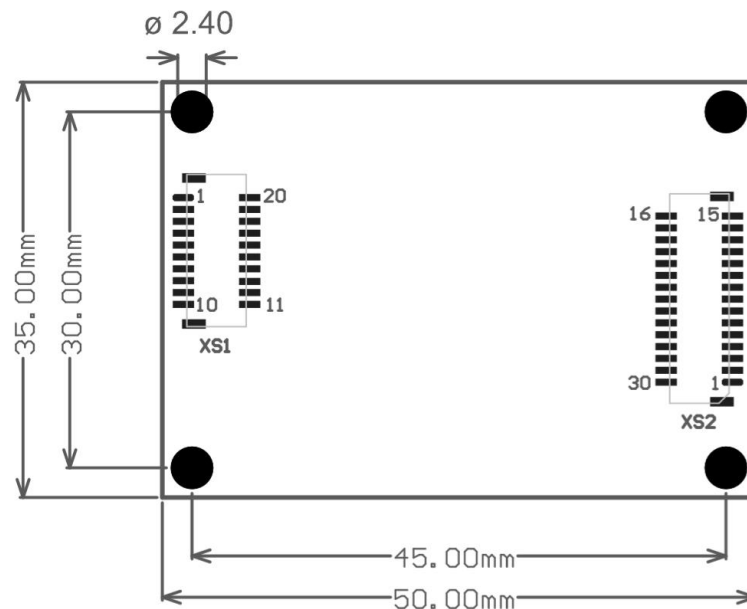
Figure 1.

- c. Weight: $\leq 0.5\text{kg}$.
- d. Appearance: The surface treatment of lens structural components adopts oxidation blackening (special parts such as trimming rings and motor gears do not require oxidation); The outer surface of optical lenses and structural components should be flat and smooth, without obvious defects such as scratches, cracks, sand holes, bumps, etc. The appearance should be neat, the coating should be firm, and there should be no rust spots.
- e. Optical interface (guaranteed by design): suitable for LEO 640×512 15 μm F4 mid-wave cooled infrared detector.

4.2 Electrical interface

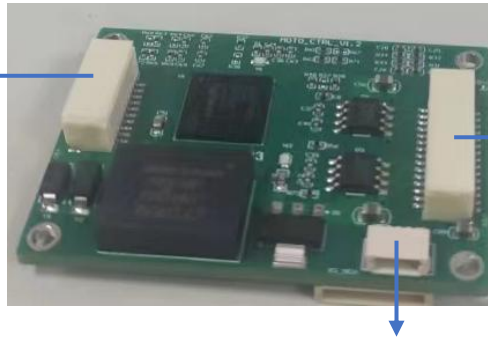
- a. Working voltage: $\text{DC}12\text{V} \pm 2\text{V}$.
- b. Communication interface: RS422 (baud rate: 19200bps).

4.3 Electrical interface



Control circuit board mechanical interfaces and sockets

Power Supply & Serial
Port 1.00-20Pin Dual Row



Motor Drive and
Control 1.00-30Pin
Double Row

Programmer Port 1.00-4Pin

Physical diagram of control circuit board

Definition of socket pins

a. XS1, Power Supply & Serial Port

SM20B-SRDS-A-G-TF	Pin number	Definition	Remarks
Power Supply	9,10	External power supply +	12V
	11,12	External power supply ground	GND
Temperature Sensor	15	Temperature sensor ground, Pin 1	
	16	Temperature sensor signal, Pin 2	
	17	Temperature sensor power supply, pin 3	
Serial Port Communication	18	Serial Port Ground	
	5	422 Serial Port RX2+	
	6	422 Serial Port TX2-	
	7	422 Serial Port RX2-	
	8	422 Serial Port TX2+	
	19	232 Serial Port RX	Control circuit board sent
	20	232 Serial Port TX	Control circuit board receive

b. XS2, Motor Drive and Control Port

SM30B-SRDS-A-G-TF	Pin number	Definition	Remarks
Focusing Motor	16	Focus -	
	17	Focus +	

	15	Encoder phase A	
	14	Encoder phase B	
	13	Encoder Phase ground	
	12	Encoder 5V	
	7	Focusing - Limit signal	Hall switches require 5V power supply, while mechanical limit switches do not
	6	Focusing + Limit signal	
	30	Focusing Limit signal ground	
	1	Focusing Limit signal 5V	
Zooming Motor	24	Zoom -	
	25	Zoom +	
	10	Encoder phase A	
	11	Encoder phase B	
	9	Encoder Phase ground	
	8	Encoder 5V	
	3	Zooming - Limit signal	Hall switches require 5V power supply, while mechanical limit switches do not
	2	Zooming + Limit signal	
	30	Zooming Limit signal ground	
	1	Zooming Limit signal 5V	

5. Operation control

Control protocol, see Appendix 1.

6. Temperature-Zoom-Focus data comparison

See Appendix 2.